

NFL Point Spreads - Solutions

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Solutions to 20 questions from the NFL Point Spreads and Game Scores module.

```
library(Hmisc)
```

Get the data set.

```
NFLPoints<-read.csv("NFLPoints.csv")
```

1. What proportion of the time does the favored team actually win the game outright?

Procedure: CI for a proportion (one sample)

Via formula

```
x=sum(NFLPoints$FavWin)
n=length(NFLPoints$FavWin)
phat=x/n
SE=sqrt(phat*(1-phat)/n)
zstar=qnorm(0.975)
ME=zstar*SE
lower=phat-ME
upper=phat+ME
```

$\hat{p} = 370/544=0.68$ with $SE=0.02$

CI for p is $0.68 \pm 1.96 \cdot 0.02 = (0.641, 0.719)$

We are 95% sure that the proportion of times the favored team wins the game is between 0.641 and 0.719.

Using an internal R function

```
prop.test(x,n,correct=FALSE)
```

1-sample proportions test without continuity correction

```
data:  x out of n, null probability 0.5
X-squared = 70.618, df = 1, p-value < 2.2e-16
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.6398066 0.7179611
sample estimates:
      p
0.6801471
```

Note that the method used for this CI is slightly different than the formula-based method above.

2. Is the chance the favorite covers the spread discernibly different from 0.50?

Procedure: Test for proportion (one sample)

$H_0 : p = 0.5$ vs. $H_a : p \neq 0.5$ where p is the proportion of games the favored team covers the spread

Via formula

```
x=sum(NFLPoints$FavCover)
n=length(NFLPoints$FavCover)
phat=x/n
SE=sqrt(phat*(1-phat)/n)
z=(phat-0.5)/SE
pvalue=2*pnorm(-abs(z))
```

$\hat{p} = 293/544=0.539$ with $SE=0.0214$

$z=(0.539 - 0.5)/0.0214=1.81$, $p\text{-value}=0.071$

At a 5% significance level we do not have convincing evidence that the proportion of times the favorite covers the spread is different from 0.50.

Using an internal R function

```
prop.test(x,n, correct=FALSE)
```

1-sample proportions test without continuity correction

```
data:  x out of n, null probability 0.5
X-squared = 3.2426, df = 1, p-value = 0.07174
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.4965875 0.5800770
sample estimates:
      p
0.5386029
```

Note that the R `prop.test` function uses a chi-square statistic, rather than a z-statistic, but the square root of that chi-square statistic should match the z-statistic up to the $\pm$ sign.

3. How different is the average point spread when the favored team is playing at home as compared to the favorite playing on the road?

Procedure: CI for a difference in means (two samples)

```
(HomePts<-t.test(Pts~HomeDog, data=NFLPoints))
```

Welch Two Sample t-test

```
data:  Pts by HomeDog
t = 3.7048, df = 503.38, p-value = 0.0002349
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 0.4430565 1.4435015
sample estimates:
mean in group 0 mean in group 1
    5.119335      4.176056
```

The difference in means is $\bar{x}_H - \bar{x}_A = 5.119 - 4.176 = 0.943$ and the 95% confidence interval for the difference in means is (0.44,1.44)